Project 1: AI Programming Foundations — Module Summary

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Overview

This report documents the data workflow built for the Road Traffic Accident Severity (RTA) dataset (Kaggle: Road Traffic Accidents), which contains 12,316 records of road traffic accidents described by 32 driver, vehicle, road, and environmental fields. The accompanying notebook, *data_workflow.ipynb*, ingests the raw CSV file, cleans missing and inconsistently formatted fields, explores severity patterns through grouped summary statistics and correlation analysis, and visualizes when and under what conditions accidents occur. This project is the first of seven capstone projects in a broader Advanced Driver-Assistance Systems (ADAS) and road-scene perception theme, in which this tabular dataset establishes a baseline understanding of accident risk factors before later projects move into deep learning and generative modeling on traffic-sign imagery.

Dataset Description

The RTA dataset records 12,316 individual road traffic accidents, each described by 32 columns spanning driver attributes (age band, sex, educational level, driving experience), vehicle attributes (type, ownership, service year, defects), roadway and environmental conditions (road surface type and condition, light conditions, weather conditions, junction type), and outcome variables (number of vehicles involved, number of casualties, casualty severity, and overall accident severity). The target variable of primary interest, *Accident_severity*, is a three-level categorical field (Slight Injury, Serious Injury, Fatal injury). Most fields are categorical (object dtype), with a small number of numeric counts (*Number_of_vehicles_involved*, *Number_of_casualties*); several categorical fields (e.g., *Defect_of_vehicle*, *Service_year_of_vehicle*, *Work_of_casuality*) contain substantial missing data, which the workflow addresses explicitly during cleaning.

Workflow Description

The notebook follows a linear ingestion-to-insight pipeline. First, the raw CSV is loaded into a pandas DataFrame and inspected with `df.head()`. Next, two cleaning functions are applied: `fill_missing_categorical`, which fills missing values in all object-typed columns with an explicit "Unknown" label, and `extract_hour_from_time`, which parses the raw *Time* strings (e.g., "17:02:00") into an integer *Hour_of_day* feature. A single EDA function, `summarize_by_severity`, then computes mean values of the numeric columns grouped by *Accident_severity* and the correlation matrix between numeric features. Finally, three visualizations are produced with Matplotlib/Seaborn to examine the severity distribution, the time-of-day pattern of accidents, and the relationship between weather conditions and casualty counts, followed by a closing narrative summary of findings.

Key Decisions and Assumptions

Several design decisions shaped this workflow. First, missing categorical values were filled with an explicit "Unknown" placeholder rather than being dropped, preserving the full 12,316-row sample for exploratory analysis; this favors completeness over strict missing-data removal and is revisited as a limitation below. Second, the raw *Time* column was parsed into an hour-of-day integer feature because the original string format was not directly usable for grouping or plotting. Third, the EDA function was designed to return both grouped means and a correlation matrix from a single, reusable, documented call, keeping analysis logic separate from presentation (plotting) logic — an organization consistent with the structured, reproducible notebook practices described by Danchev (2022) in the required article on reproducible data science with Python. Figure 1 (Distribution of Accident Severity) was chosen first because understanding class balance in the target variable is a prerequisite for any later modeling decision. Figure 2 (Number of Accidents by Hour of Day) was chosen to test whether accident frequency tracks traffic density. Figure 3 (Average Number of Casualties by Weather Condition) was chosen to examine whether environmental conditions are associated with injury severity. Structuring the cleaning and EDA steps as small, named, documented functions rather than inline, one-off code also follows general best practice for organizing tabular data workflows into clear, well-defined variables and transformations, as emphasized in work on tidy, structured tabular data (Wickham, 2014).

Results and Interpretation

Figure 1 shows that *Accident_severity* is highly imbalanced: Slight Injury accounts for the large majority of records (approximately 85%), Serious Injury for a much smaller share, and Fatal injury for only about 1.3% of records.

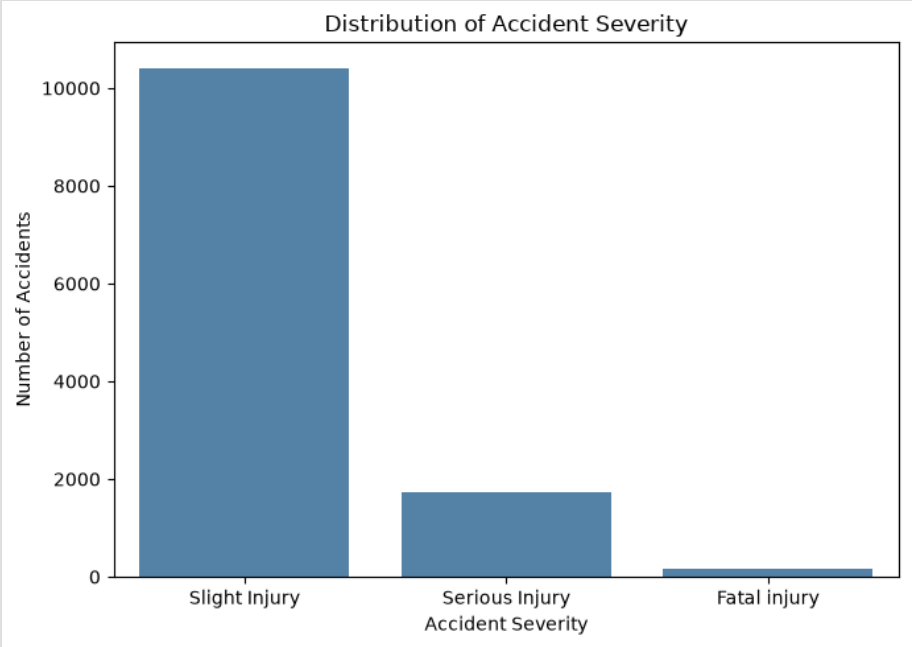


Figure 1. Distribution of Accident Severity — count of accidents per severity class.

Figure 2 shows that accident counts rise steadily through the day and peak sharply in the late afternoon, around 17:00, consistent with higher traffic volume during the evening commute, before falling to their lowest levels in the early morning hours.

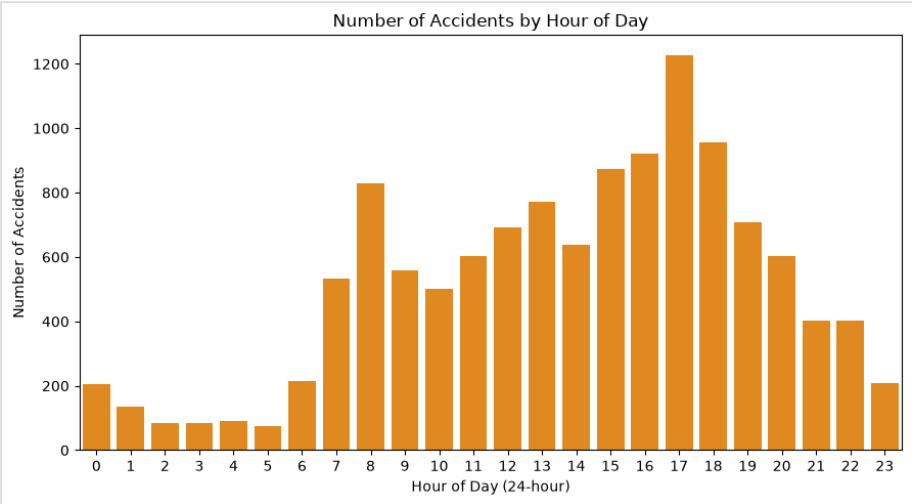


Figure 2. Number of Accidents by Hour of Day — accident counts across the 24-hour cycle.

Figure 3 shows that average casualty counts are relatively stable (roughly 2 to 2.3) across most weather conditions, with a small number of rarer categories (e.g., Windy, Snow) showing modestly higher averages; because these categories contain very few observations relative to Normal conditions, this difference should be interpreted cautiously rather than as strong evidence of a weather effect.

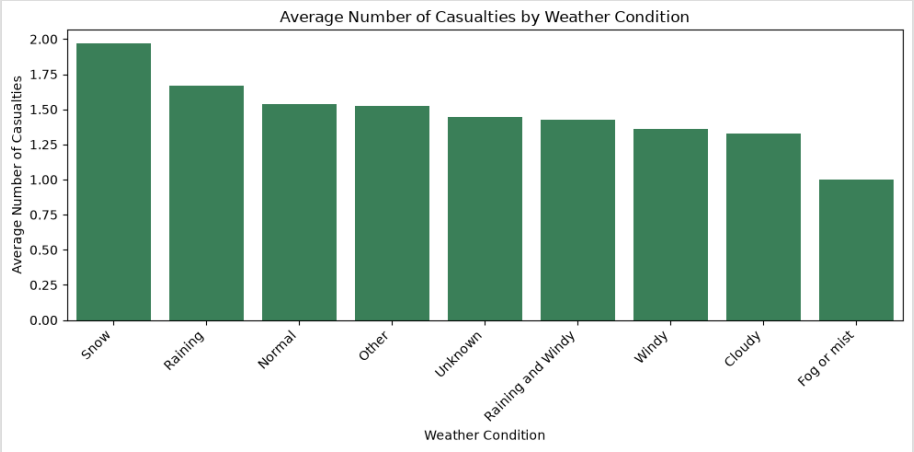


Figure 3. Average Number of Casualties by Weather Condition.

Taken together, these results suggest that temporal exposure (time of day) is a clearer driver of raw accident frequency in this dataset than weather condition is of casualty severity.

Responsible Practice

This dataset carries several bias and data-quality risks that should be acknowledged before any downstream use. First, the severe class imbalance in *Accident_severity* means a naive model could achieve high accuracy by simply predicting "Slight Injury" for every case while failing to identify the rare but most consequential Fatal injury outcomes; any future modeling work on this dataset should use severity-aware metrics (e.g., recall on the minority class) rather than raw accuracy. Second, several fields had large amounts of missing data (e.g., *Defect_of_vehicle*, *Service_year_of_vehicle*), and filling these with "Unknown" risks conflating genuinely missing data with legitimate "not applicable" cases (e.g., no vehicle defect); if this ambiguity is not documented, later analyses could draw incorrect conclusions about how common such factors actually are. Third, this dataset's regional provenance means the recorded patterns may not generalize to other geographies, road networks, or reporting standards, so any claims drawn from this analysis should be scoped explicitly to the population the data represents. Mitigations applied

here include explicitly labeling imputed values rather than silently imputing a mode or mean, and reporting class proportions directly in the results above rather than only reporting aggregate statistics that could mask the imbalance.

Reproducibility

The workflow is designed to be reproducible end to end. All Python dependencies are pinned via `pip freeze` into *requirements.txt*, so the exact package versions used to build this notebook can be reinstalled with `pip install -r requirements.txt`, directly supporting the reproducible-workflow guidance described by Danchev (2022). Version control is managed with Git: the project was initialized on a *main* branch holding the dataset, environment configuration, and README, and all cleaning, EDA, and visualization work was developed on a separate *data-cleaning-eda* branch before being merged back via a pull request, keeping exploratory notebook changes isolated from the stable project baseline until reviewed.

Future Integration Reflections

Moving from this exploratory workflow to training a machine learning model would require additional steps not needed here: categorical features would need to be encoded (e.g., one-hot or ordinal encoding) rather than left as raw strings, the "Unknown" placeholder values would need a documented imputation strategy rather than being treated as an ordinary category, and the data would need to be split into training, validation, and test sets with stratification on *Accident_severity* to preserve class proportions across splits. Before training a neural network specifically, it would also be important to normalize or scale numeric features, verify there is enough data per class given how few Fatal injury examples exist relative to the roughly 10,000 available for Slight Injury, and choose a loss function (such as class-weighted cross-entropy) that accounts for the severe class imbalance identified in Figure 1. Agentic automation could usefully assist later stages of this pipeline, such as automatically flagging newly ingested data for schema drift or unexpected missingness rates before cleaning runs, or routing low-confidence severity predictions to a human reviewer; however, agentic automation would be risky if applied to the labeling or correction of ambiguous "Unknown" values without human oversight, since an agent could silently introduce bias by guessing values for the rare, safety-critical Fatal injury cases this dataset is least able to describe reliably.

Sources and Citations

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, *5*(56), 156. https://doi.org/10.21105/jose.00156

Wickham, H. (2014). Tidy data. *Journal of Statistical Software*, *59*(10). https://doi.org/10.18637/jss.v059.i10